

EDUCATION

Harvard Medical School | Boston, MA

MS in Biomedical Informatics | GPA: 3.97

Dec 2024

- Relevant Coursework:** ML for Computational Biology (MIT), Single Cell Analysis, Health Law & Policy (HLS), Affective Computing (MIT Media Lab), Building Successful Enterprises (HBS), Rare Genetic Disease Diagnostics
- Activities:** Harvard Biotech Club, Harvard Innovation Labs, Harvard Graduate Consulting Club, Camp Kesem

Northwestern University | Evanston, IL

BA in Computational & Systems Biology, Data Science, and Linguistics | GPA: 3.93 (Dean's List 8/9)

Jun 2023

- Honors Program in Medical Education (**HPME**): 7-year BA/MD program with Feinberg School of Medicine
- Relevant Coursework:** Data Science I-III, Data Visualization, Text Processing, Bioinformatics, Biostatistics

PROFESSIONAL & RESEARCH EXPERIENCE

Co-Founder & CIO | Bond Health | New York, NY

Jan 2024 – Present

- Build proprietary LLM-based platform to improve the efficiency of clinical trial matching and patient screening
- Direct HIPAA-compliant pilot integrations with clinical sites, enabling scalable personalized screening workflows
- \$150K+ total investments from CEAS Investments, Harvard Innovation Labs, MIT Sandbox, and NSF I-Corps

ML Researcher | Center for Precision Psychiatry, Mass General Hospital | Boston, MA

Dec 2023 – Dec 2024

- Developed a metabolomics framework for an accessible/severity-stratified approach for MDD early detection
- Conducted multivariable regression models revealing lipid-driven metabolic patterns reflecting MDD phenotype
- Engineered mutual information network pipelines to isolate informative metabolic biomarkers from 250+ features

Data Scientist Internship | Behavior | Remote

Sep 2023 – Dec 2023

- Implemented RNNs and random forests on 100+ parameter, 200k physiological datasets to predict cravings
- Leveraged label propagation algorithms for improved scikit-learn clustering (15% gain in model precision)
- Improved Cypher algorithm efficiency for accelerated real-time anomaly detection in patient wearable devices

STRATEGY & CONSULTING EXPERIENCE

Associate Consultant Internship | H2Ok Innovations | Boston, MA

Oct 2023 – Dec 2023

- Scaled application of artificial IoT-enabled industrial liquid optimization across top-tier Fortune 500 enterprises
- Project manager:* conducted strategic market and industry sensitivity analyses for a \$14 billion market entry
- Enhanced revenue generation by 25% through tailored, client-specific pricing models from competitive analyses

Strategy Consultant | Mercurial-AI | Chicago, IL

Mar 2023 – Aug 2023

- Developed and implemented strategic operational changes that successfully secured \$300K+ in seed capital
- Project manager:* overlooked deep ensemble/casual framework personalizing breast cancer treatments
- Conducted competitive landscape and market sizing analyses to inform AI-driven oncology product positioning

PUBLICATIONS & PROJECTS

Evaluating Pain Assessment and Management Disparities in Acute Pancreatitis – MIT

- Exposed racial and gender inequities in opioid prescribing, treatment delays, and pain relief outcomes in the ED
- Analyzed ~15k clinical records to design metrics capturing pain documentation and treatment response patterns

A Medico-Legal Approach to Subjective Pain Testimony in Disability Adjudication – Harvard Law School

- Uncovered systemic mishandling of subjective pain testimony by administrative law judges in disability rulings
- Proposed coalition-led reforms leveraging CMS quality surveys to standardize pain documentation for legal use

SERBP1: Exploiting RNA-Binding Protein-Mediated PAI-1 Inhibition – Feinberg School of Medicine

- Developed a transgenerational epigenetic method of prion-mediated PAI-1 destabilization for cardiac fibrosis
- Compared gene expression patterns with and without SERBP1 treatment *in vitro* via PCA on RNA-Seq data in R

AWARDS

Regeneron Science Talent Search Scholar

\$10,000 HPME and Weinberg Northwestern Research Grants

MIT \$100K Entrepreneurship Competition Semifinalist

\$5,000 Thermofisher Scientific Scholarship Winner

SKILLS

Languages:

Python, R, SQL, C++, Cypher, HTML, JavaScript, MATLAB, Bash

Technologies:

PyTorch, TensorFlow, Git, NLP, API Development, Docker, AWS, FHIR/HL7, Linux, Figma

Additional Skills:

Spanish (Intermediate), Arabic (Intermediate), Photography